Lab: Visualization – How Many Cluster Are There?

How well does the data cluster?

- four step exercises

How_Many_Cluster_Are_There_I
How_Many_Cluster_Are_There_II
How_Many_Cluster_Are_There_III
How_Many_Cluster_Are_There_IV

- each step:
  - we discuss the problem set to be solved
  - you ask questions
  - you will **work in groups in breakout rooms**
  - after time is up: return to the **main room** and **discuss the solution**
  - you will **get the sample solution** and **compare it to your own solution**
  - material for **next step** will be pushed to **277A repo**
  - repeat: next step

- You don’t need to turn anything in. You will have the complete solution at the end.

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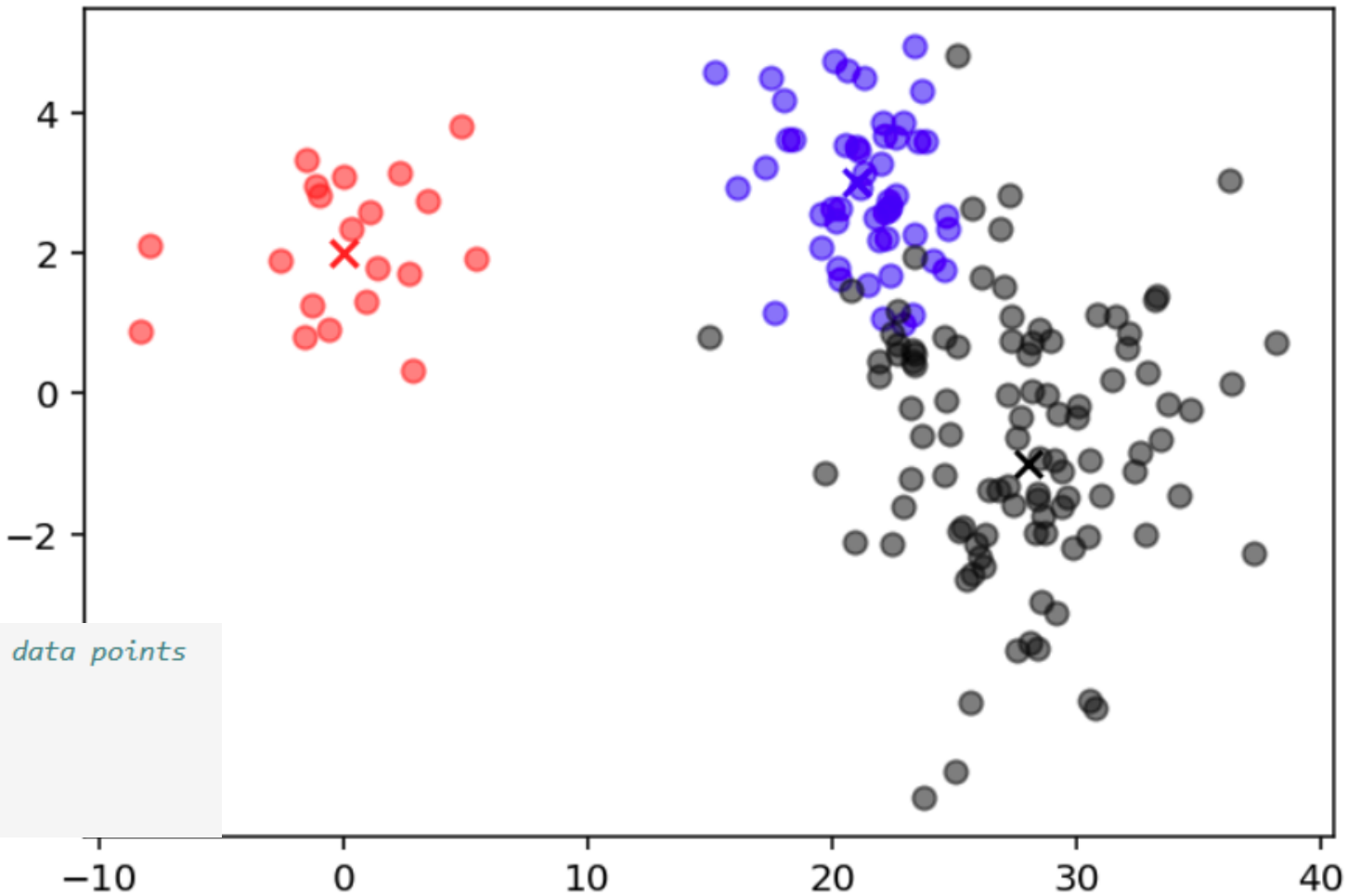
I:

2.1) Generate three 2D cluster:

- Write a small function using `def` that generates three cluster from `np.random.normal`, each with two coordinates.
- The function should create a scatter plot of the three cluster including highlighting the mean (see example below).
- experiment with different cluster sizes N, values for μ and σ.




```
def ThreeCluster(N:   tuple = (20, 50, 100), # number of data points
                 Mux: tuple = ( 1,  5,   0), # mu for x
                 Muy: tuple = ( 2,  3,  -1), # mu for y
                 Stx: tuple = ( 3,  2,   4), # std for x
                 Sty: tuple = ( 1,  1,   2)):# std for y
```

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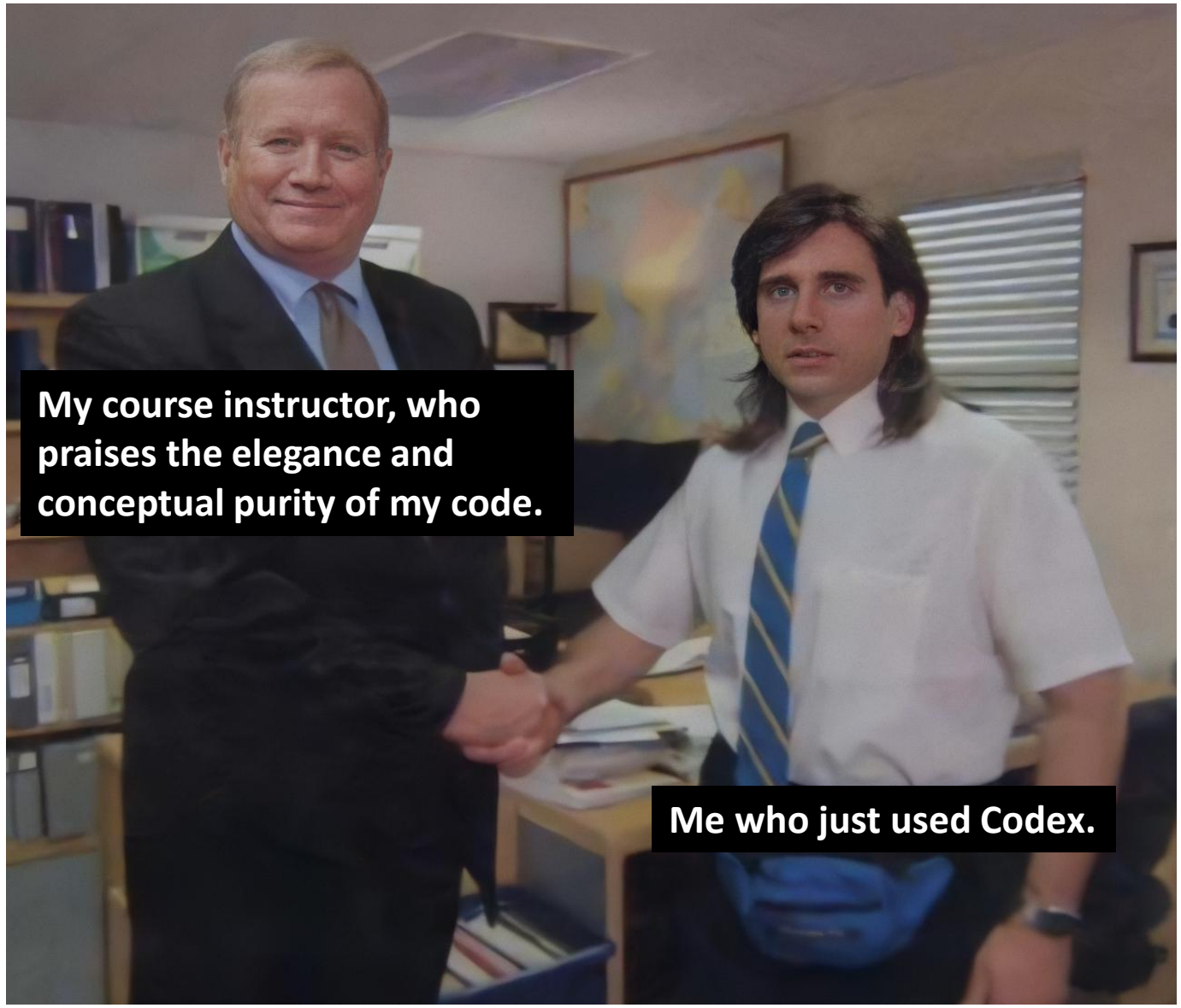
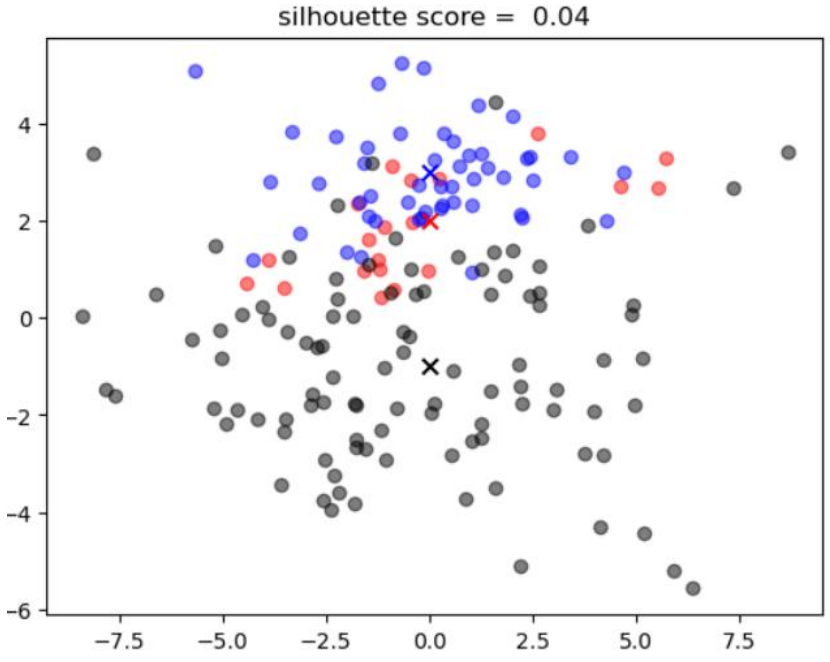
Enjoy :)

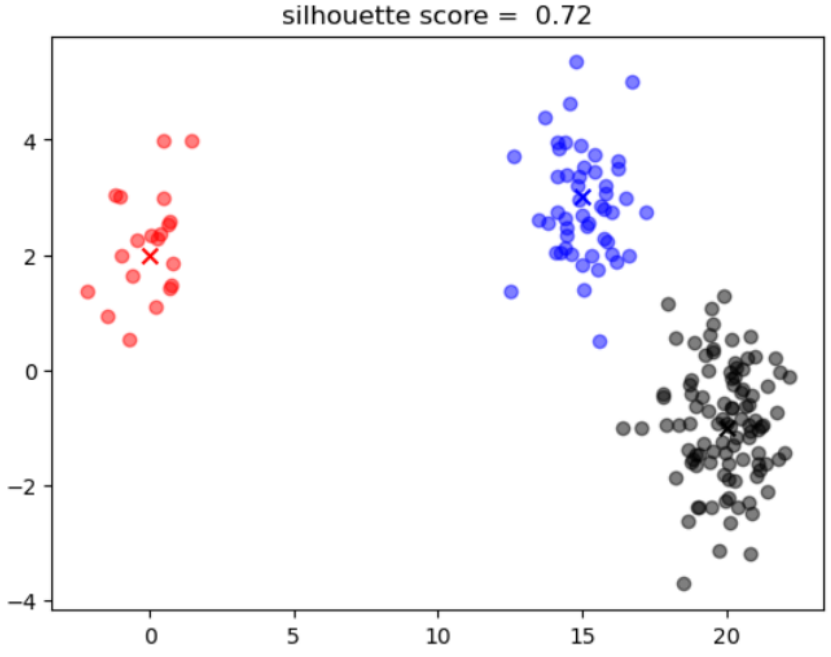


How well does the data cluster?

II: 2.2) How well do they cluster?

- Modify `ThreeCluster` such that it calculates the silhoutte score ψ_{tot} of the dataset via `score = silhouette_score(X, Labels, metric = 'euclidean')` and shows the silhoutte score ψ_{tot} in the plot title. Note, that you know the labels already, since you created the cluster yourself. `X` is the entire dataset containing all data points.
- Next, run `ThreeCluster` systematically for different values for μ and σ (i.e. vary μ and σ one at a time, therefore increasing the distances between the cluster, making them more scattered etc). How does that influence ψ_{tot}? For which values of μ and σ does ψ_{tot} start to change?





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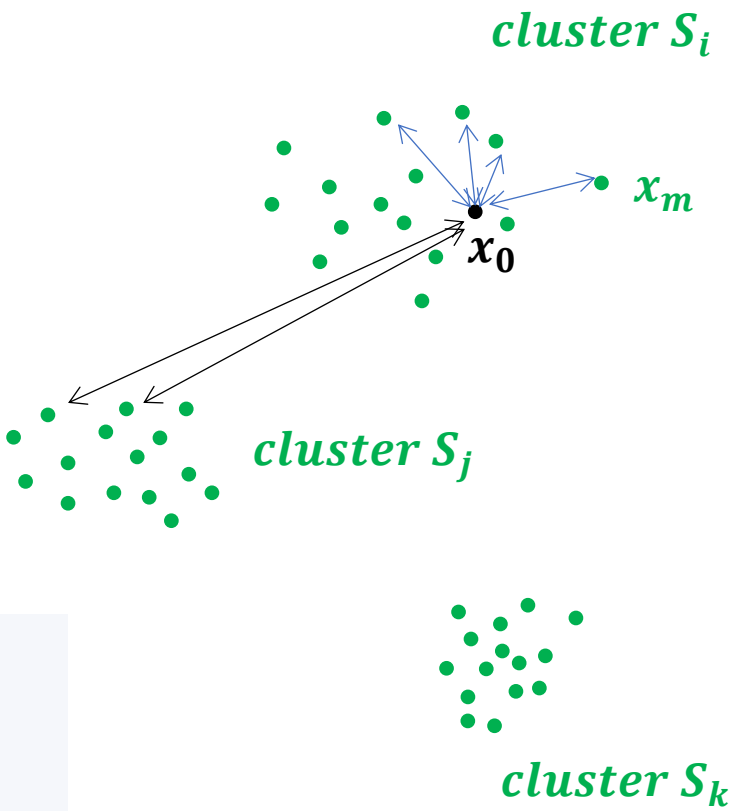
- distance $\boldsymbol{d_1}$ of a data point x_0 to ***its assigned cluster*** $\boldsymbol{S_i}$
  vs distance $\boldsymbol{d_2}$ to ***closest cluster (here*** $\boldsymbol{S_j}$***)***

$$\boxed{\Psi(x_0) = \begin{cases} 0 & if\ d_1 = 0 \\ \dfrac{d_2 - \boldsymbol{d_1}}{max[\boldsymbol{d_1}; d_2]} & \end{cases}}$$



- average over all points → ψ_{tot}

if	$\psi_{tot} = 0.75 \ldots 1.00$	→ well clustered
	$\psi_{tot} = 0.50 \ldots 0.75$	→ medium clustered
	$\psi_{tot} = 0.25 \ldots 0.50$	→ poorly clustered
	$\psi_{tot} < 0.25$	→ data has no structure

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How well does the data cluster?

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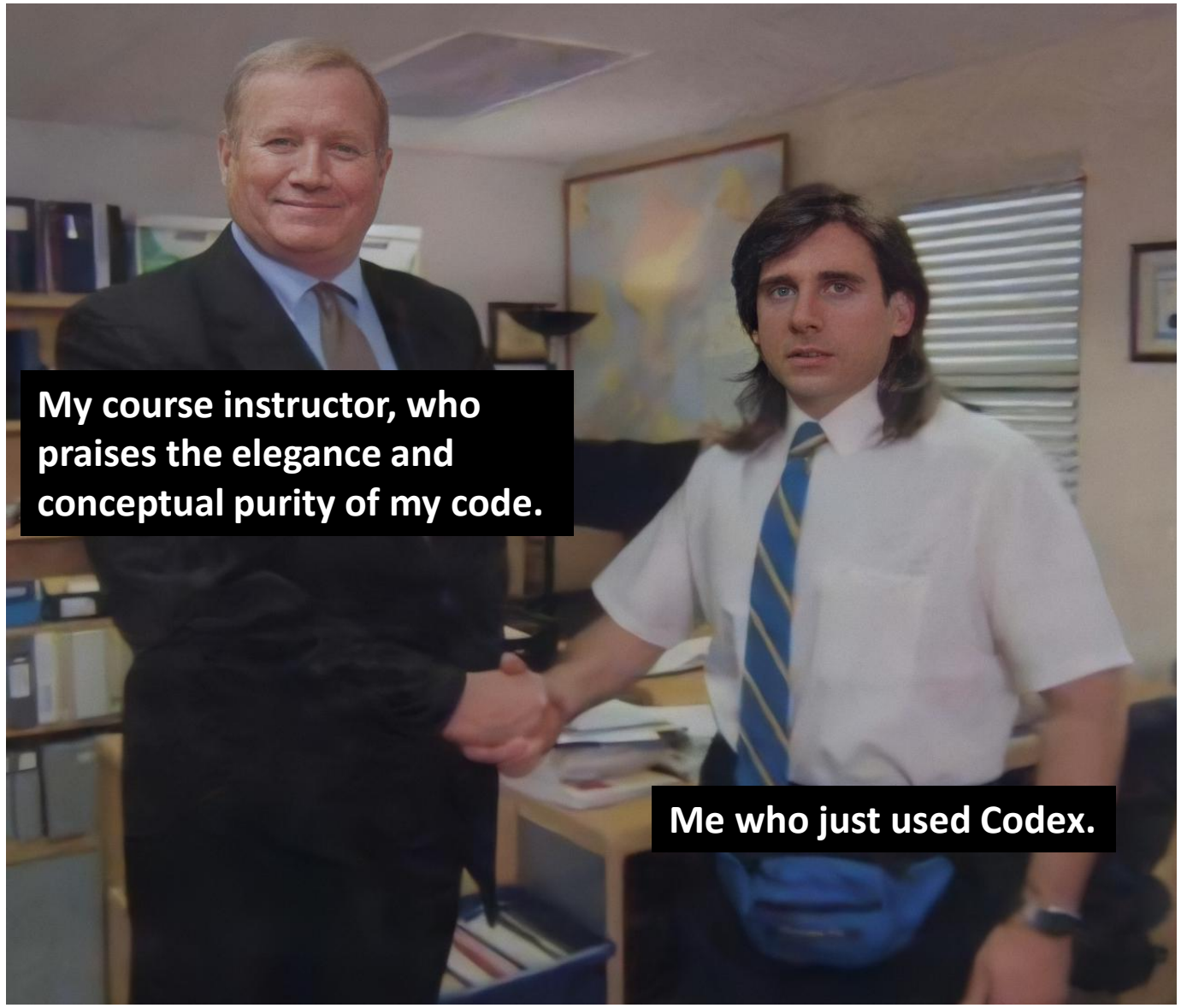

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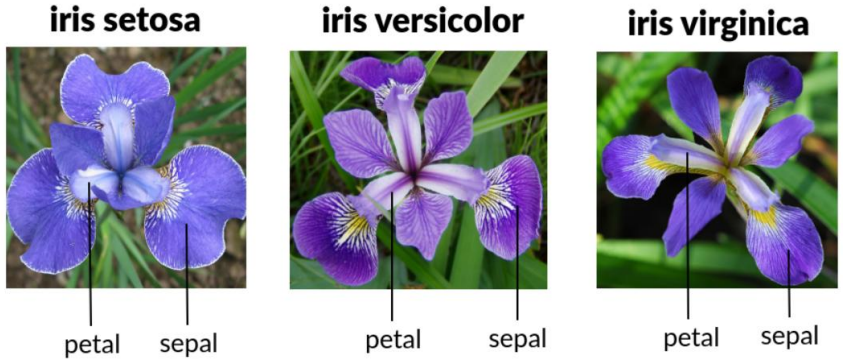
III:




```
:Number of Instances: 150 (50 in each of three classes)
:Number of Attributes: 4 numeric, predictive attributes and the class
:Attribute Information:
    - sepal length in cm
    - sepal width in cm
    - petal length in cm
    - petal width in cm
    - class:
            - Iris-Setosa
            - Iris-Versicolour
            - Iris-Virginica
```

4D

3 classes = 3 cluster?

```
:Summary Statistics:

============== ==== ==== ======= ===== ====================
                Min  Max   Mean    SD   Class Correlation
============== ==== ==== ======= ===== ====================
sepal length:   4.3  7.9   5.84   0.83    0.7826
sepal width:    2.0  4.4   3.05   0.43   -0.4194
petal length:   1.0  6.9   3.76   1.76    0.9490  (high!)
petal width:    0.1  2.5   1.20   0.76    0.9565  (high!)
============== ==== ==== ======= ===== ====================
```

The Iris dataset has 4 features, so it cannot be visualized in a single 2D plot. Create scatter plots for every pair of features (for example: feature 1 vs feature 2, feature 1 vs feature 3, etc.), similar to the example from the PCA lecture. What do you observe? Do they cluster well?

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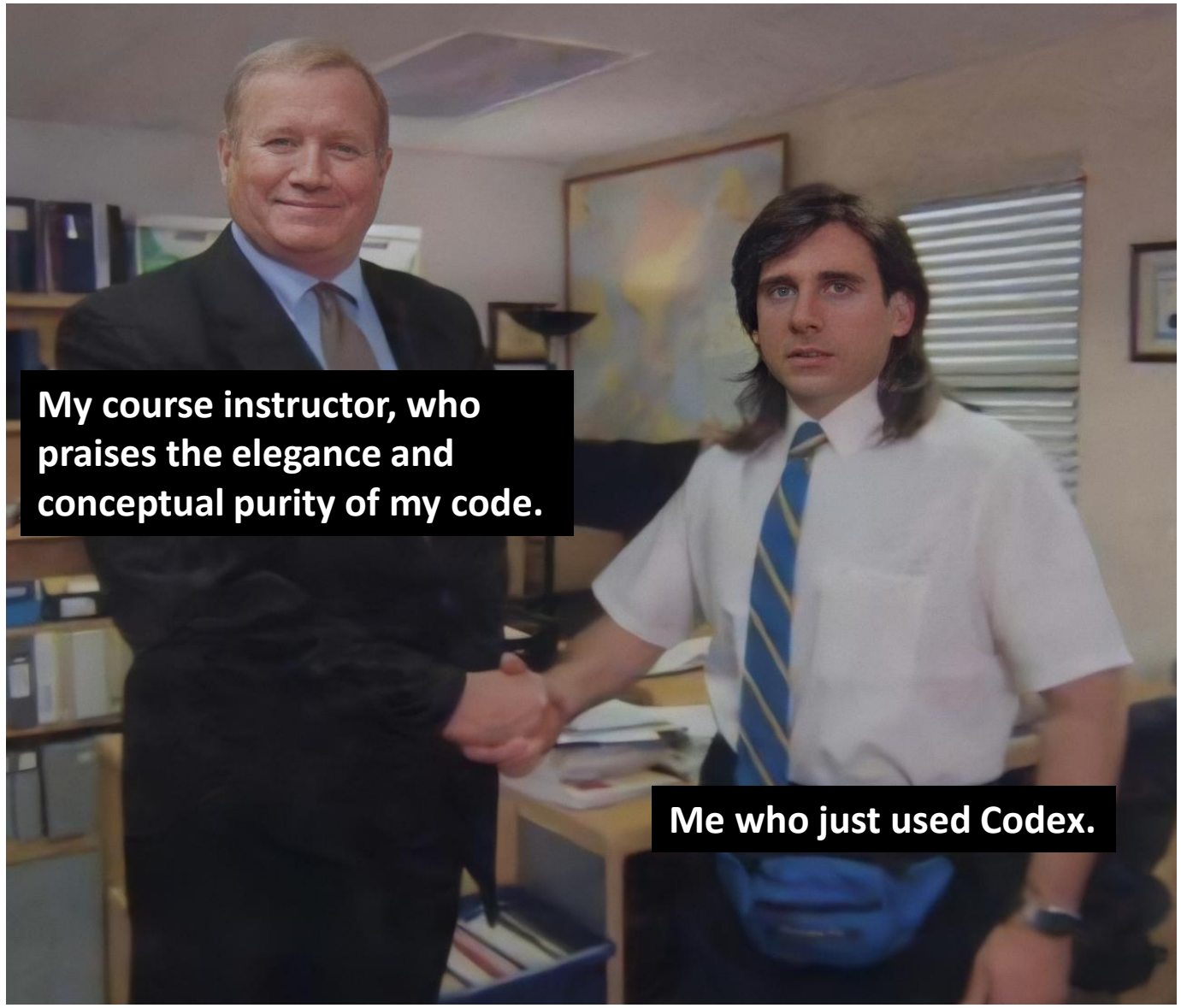


Lab: Visualization – How Many Cluster Are There?

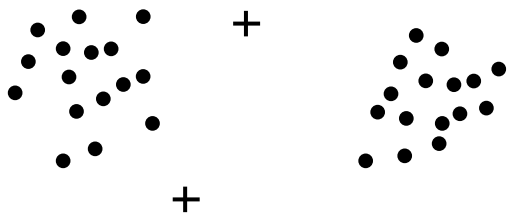
How well does the data cluster?

IV: k-means

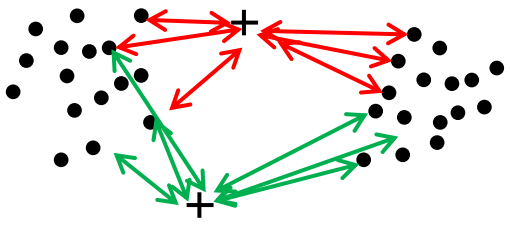
K	: number of cluster
N	: number of observations
μ_k	: barycenter of the cluster

$$r_{n,k} = \begin{cases} 1, & if\ k = argmin_j \|x_n - \mu_j\|^2 \\ 0, & else \end{cases}$$

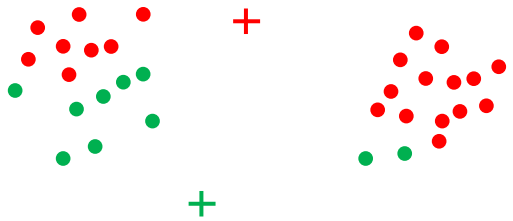
$$J = \sum_{n=1}^{N} \sum_{k=1}^{K} r_{n,k} \ \|x_n - \mu_k\|^2$$



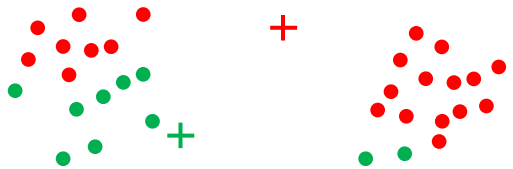
a) assign *k* means randomly



b) calculate *distance* from each point to each mean



c) assign each point to its closest mean



d) update the means accordingly

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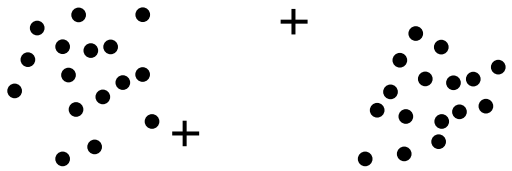
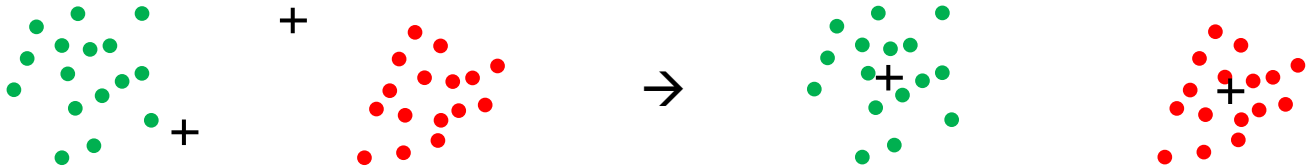


e) go back to b)



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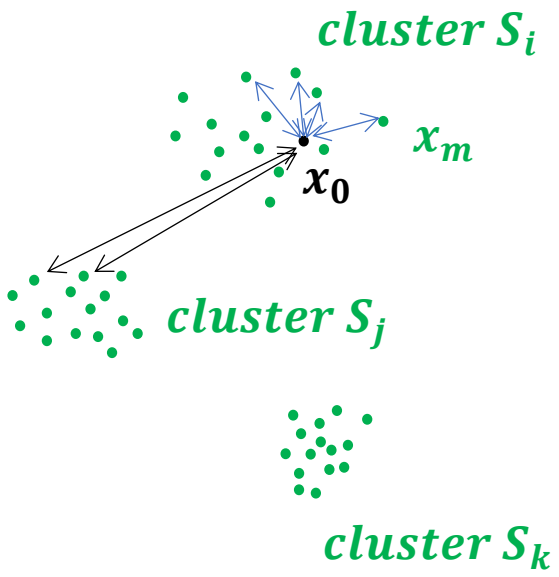

silhouette Ψ

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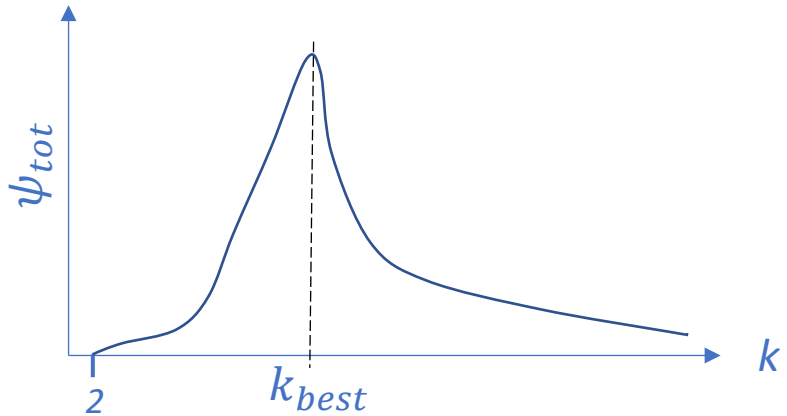
ideal world →



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IV: k-means

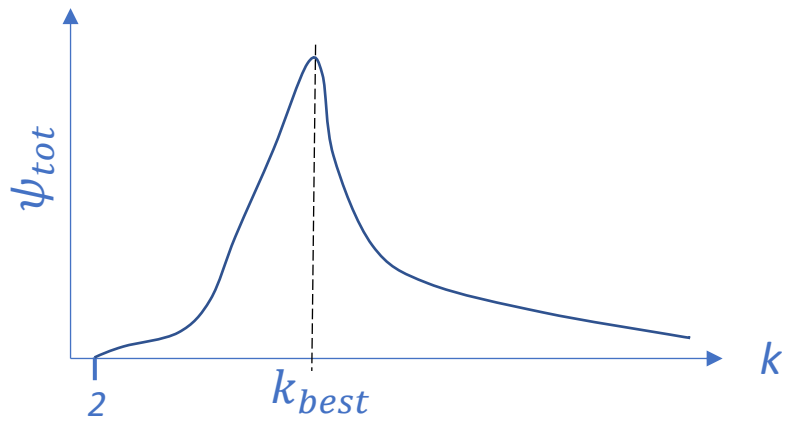
2.4) Running a simple cluster algorithm.

Assuming, we wouldn't know how many classes are there, we would run a clustering algorithm, assuming a certain number of cluster k. This clustering algorithm returns the labels and based on the labels, we can calculate the silhoutte score. The idea is that the best clustering, hence **best k, will return the highest silhoutte score** and we thereby find k.

- Run a so-called k-means clustering via

```
my_model = KMeansClusterer(k, distance = distance_metric(type_metric.EUCLIDEAN), repeats = 25, avoid_empty_clusters = True)
Labels   = my_model.cluster(X, assign_clusters = True)
```

for different $k = 2, \ldots 15$ and calculate ψ_{tot}. Please plot the mean silhouette score vs number of clusters. For which k do you get the highest ψ_{tot}?



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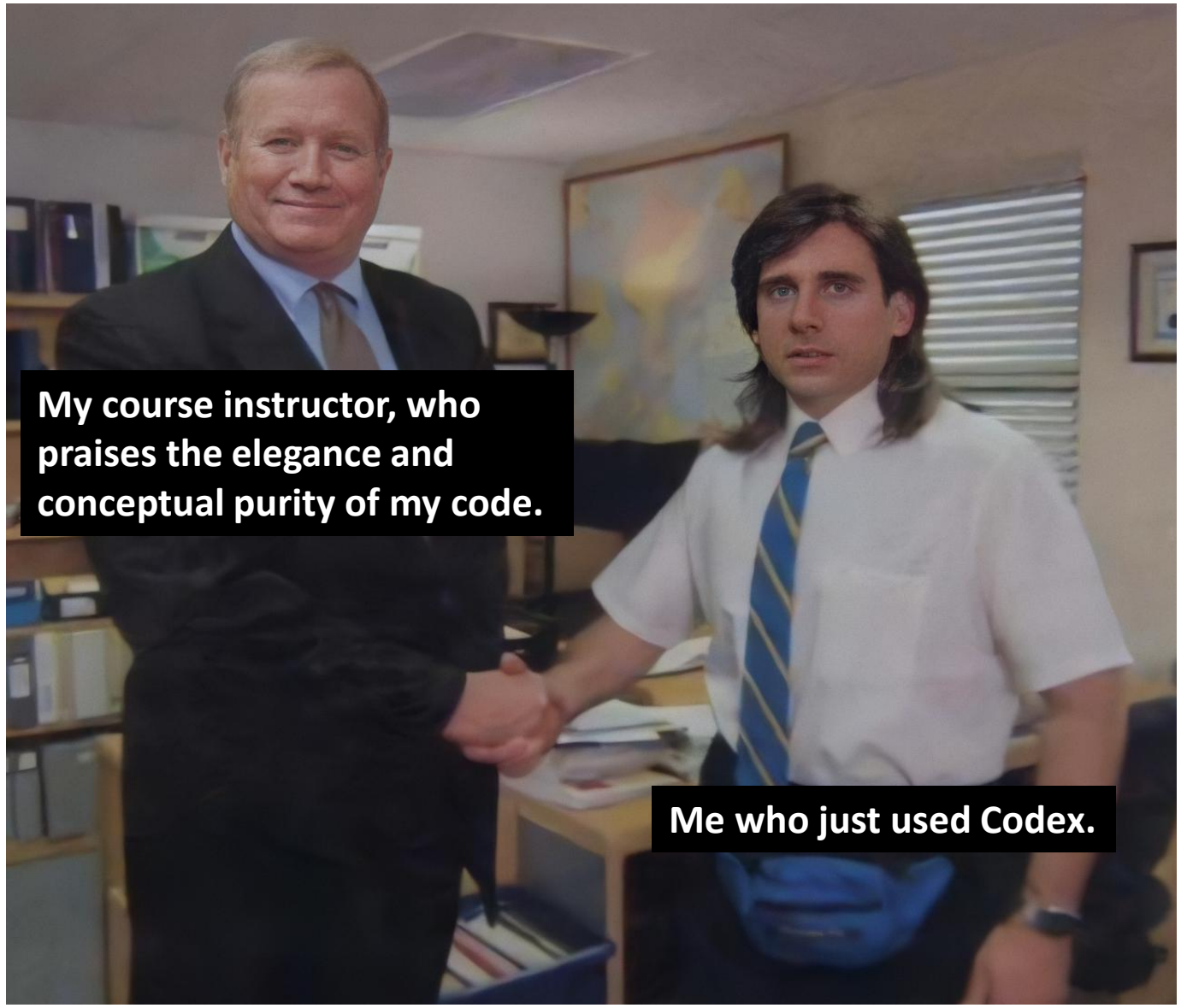
Enjoy :)